PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

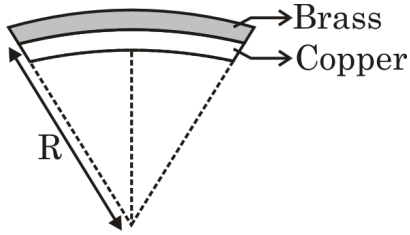
- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : –2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

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1. A bimetallic strip is formed out of two identical strips, one of copper and other of brass. The coefficient of linear expansion of the two metals are α_c and α_B respectively. On heating, the temperature of the strip increases up by ΔT and the strip bends to form an arc of radius of curvature R. Then R is :-



(A) Proportional to ΔT.

(B) Inversely proportional to ΔT.

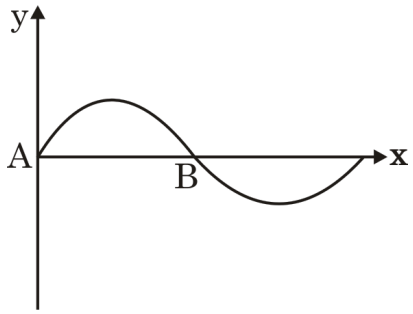
(C) Proportional to $|\alpha_B - \alpha_C|$.

(D) Inversely proportional to $|\alpha_B - \alpha_C|$.

Ans. B, D

---

2. The tension in a stretched string fixed at both ends is changed by 2%, the fundamental frequency is found to get changed by 15 Hz. Select the correct statement(s).



(A) Wavelength of fundamental frequency does not change.

(B) Velocity of propagation of wave changes by 2%.

(C) Velocity of propagation of wave changes by 1%.

(D) Original frequency is 1500 Hz.

Ans. A, C, D

3. Two pulses travelling on same string are described by equations $y_1 = \frac{5}{(3x-4t)^2+2}$ and $y_2 = \frac{-5}{(3x+4t-6)^2+2}$. Mark out correct statements.

(A) At $t = \frac{4}{3}$s the two waves cancel each other everywhere.

(B) At $t = \frac{3}{4}$s the two waves cancel each other everywhere.

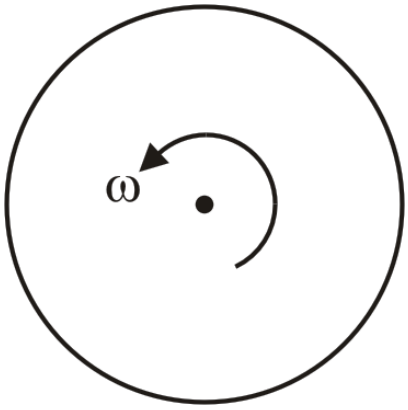
(C) At x = 1 the two waves cancel each other at all times.

(D) At all points two waves cancel each other for all values of t.

Ans. B, C

SPACE FOR ROUGH WORK

4. A wire of mass M, length L and cross-section A is shaped into a ring. It is placed on smooth horizontal surface and is rotated about its axis with constant angular speed ω as shown. If Young's modulus of material of wire is Y, then :-



(A) Tension developed in the wire is $\frac{M\omega^2 L}{\pi^2}$.

(B) Elongation produced in the wire is $\frac{M\omega^2 L^2}{4\pi^2 Y A}$.

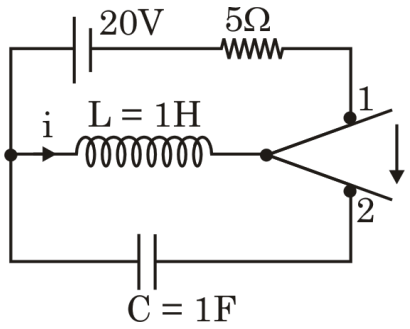
(C) Stress developed in the wire is $\frac{M\omega^2 L}{4\pi^2 A}$.

(D) Strain developed in the wire is $\frac{M\omega^2 L}{\pi^2 Y A}$.

Ans. B, C

SPACE FOR ROUGH WORK

5. In the circuit shown, the switch has been in position 1 for a long time. Now the switch is shifted to position 2. If this instant is taken as t = 0, then at time t = $\pi/2$:-



(A) Current through the inductor is zero.

(B) Current through the inductor is 4 A.

(C) Charge on the capacitor is 4 C.

(D) Potential difference across the capacitor is 4 V.

Ans. A, C, D

6. Three simple harmonic motions about the same point in the same direction having the same amplitude a and same period are superposed. If the 3rd SHM leads the second by 45° and the second leads the first by same phase angle, then :-

(A) The resultant amplitude is $(1 + \sqrt{2})a$.

(B) The phase of the resultant motion relative to the first is 90°.

(C) The energy associated with the resulting motion is $(3 + 2\sqrt{2})$ times the energy associated with any single motion.

(D) The resulting motion is not simple harmonic.

Ans. A, C

SPACE FOR ROUGH WORK

SECTION-I (ii) : (Maximum Marks: 12)

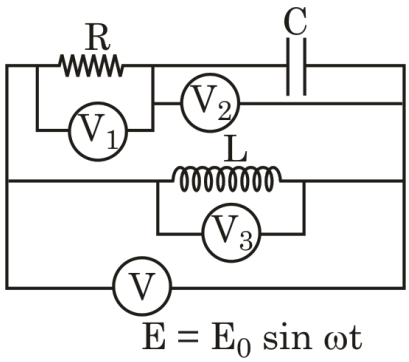
- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : –1 In all other cases.

SPACE FOR ROUGH WORK

Paragraph for Questions 7 and 8

Consider AC circuit given below



7. ω for which average power given by source is maximum, would be :-

(A) $\frac{1}{\sqrt{LC - R^2C^2}}$

(B) $\frac{1}{\sqrt{LC}}$

(C) $\frac{1}{\sqrt{\frac{L}{C} - R}}$

(D) $\frac{1}{R}\sqrt{\frac{L}{C}}$

Ans. A

8. If reading of v_2 and v_3 is 30 volts and 50 volts respectively, then reading of v_1 would be :-

(A) 20 V

(B) $20\sqrt{2}$ V

(C) $10\sqrt{2}$ V

(D) 40 V

Ans. D

Paragraph for Questions 9 and 10

A hollow cylinder has length L, inner radius a and outer radius b. Thermal conductivity is k. Inner and outer surfaces of the cylinder are maintained at temperatures 100°C and 0°C respectively. If kL = 11 J/s, a = 0.1 m, b = 0.3 m, and $\ell n3 = 1.1$.

9. The rate of radial heat (use $\pi = 3.14$) flow (in kJ/s) is :-

(A) 8.26

(B) 6.28

(C) 5.75

(D) 7.65

Ans. B

10. If the temperature gradient near the inner surface of cylinder is $\frac{1000n}{11}$ °C, then value of n is :-

(A) 10

(B) 15

(C) 20

(D) 25

Ans. A

SPACE FOR ROUGH WORK

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

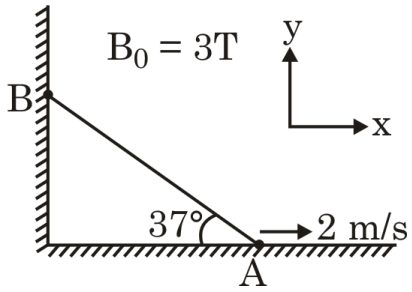
Full Marks : +3 If only the correct answer is given.
Zero Marks : 0 In all other cases

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1. A certain amount of ice is heated at a constant rate for 16 minutes. For the first 2 minutes, the temperature rises uniformly with time, then it remains constant for next 10 minutes and again the temperature rises at uniform rate for last 4 minutes. Final temperature at the end of 16 minutes (in °C), is :- [C_{ice} = 0.5 cal/g°C, C_{water} = 1 cal/g°C, L_f = 80 cal/g]

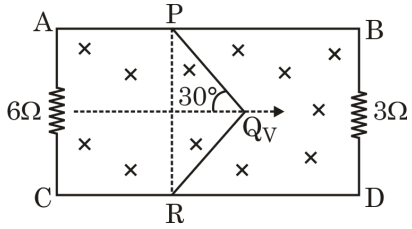
Ans. 32

2. A conducting rod of length 5 m slides in a vertical plane against two insulated perpendicular surfaces as shown. Outward uniform magnetic field B_0 = 3 tesla exists in the region. If end A of the rod is being pulled with constant speed $V_0 = 2\ m/s\hat{i}$, then find the magnitude of induced Emf (in volt) across end A and B of the rod at the instant when the rod makes an angle of $\theta = 37°$ with horizontal.



Ans. 7

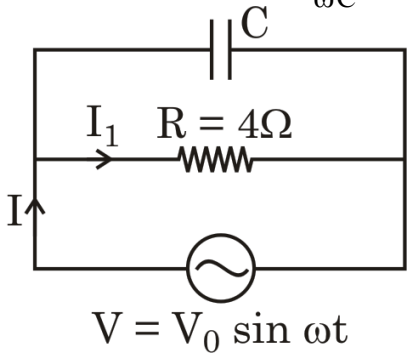
3. A conductor PQR of length 0.6 metre and with a resistance r = 1Ω moves on two parallel conducting rails CD and AB with a uniform speed V = 4 m/s normal to a uniform magnetic field of induction B = 5 T. The rails are connected by two resistance as shown in figure. If mechanical power needed for the motion is 3x watt, then find out the value of x.



Ans. 4

SPACE FOR ROUGH WORK

4. A capacitor and a resistor are joined in parallel to an AC source as shown in the figure. The phase difference between currents I and I_1 is 53°. Find the value of $\frac{1}{\omega C}$ (in SI unit).

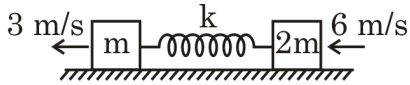


Ans. 3

5. A capacitor of capacitance 1 μF is charged to potential difference 2 V and is connected to an inductor of 1 mH. At an instant when potential difference across the capacitor is 1 V, the current in the circuit is $10^{-2}\sqrt{\frac{10x}{3}}$ ampere. Find out the value of x.

Ans. 9

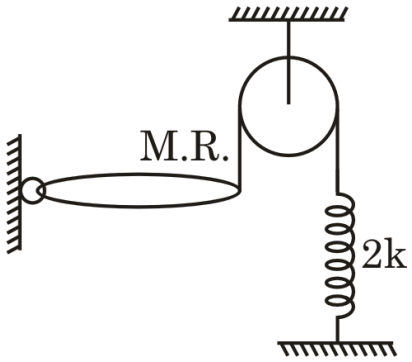
6. Consider two blocks of mass m and 2m which are connected by an ideal spring of spring constant k. Complete arrangement is placed on smooth horizontal plane. At t = O velocities 3 m/s and 6 m/s are given to m and 2m as shown in figure. If t = t_1 and t = t_2 is the first time instants (t_1 and $t_2 \neq 0$) when block of mass m have maximum velocity and 2m have minimum velocity respectively then value of $|t_1 - 2t_2|$ comes out to be $\pi\sqrt{\frac{2m}{Nk}}$, find N.



Ans. 3

SPACE FOR ROUGH WORK

7. Consider a ring of mass M = 1 kg and radius R which is connected to an inextensible string which passes over a frictionless pulley. The other end of string is connected to upper end of a massless spring having spring constant 2k (k = 25 N/cm). Ring can rotate in the vertical plane about hinge without any friction. If horizontal position of ring is in equilibrium position then time period of small oscillation of the ring comes out $T = \frac{\pi}{100}\sqrt{N}s$, find value of N.



Ans. 3

8. Two vibrating strings of the same material but length L and 2L have radii 2r and r respectively. They are stretched
under the same tension. Both the strings vibrates in their fundamental modes, the one of length L with frequency f_1 and the other with frequency f_2. Then the ratio $\left(\frac{f_1}{f_2}\right)$ is :-

Ans. 1

SPACE FOR ROUGH WORK

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- This section contains **THREE** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has Four options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
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  Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

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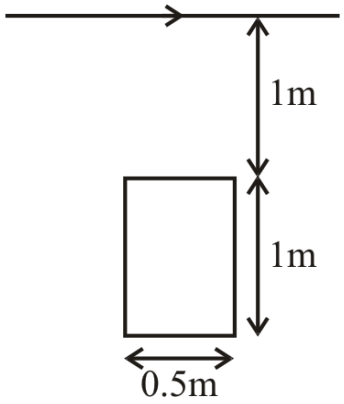
  Negative Marks : –2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

SPACE FOR ROUGH WORK

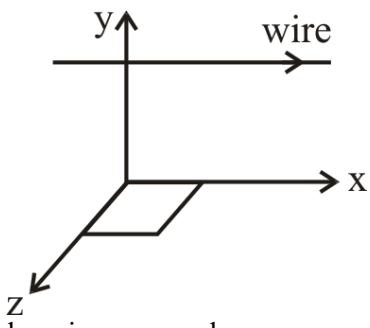
Paragraph for Questions 1 and 2

People waiting near the tracks of an electrically driven train have sometimes noticed a tingling if they touch a conducting object, such as a pipe, connected to the ground. What causes the tingling? The reason is that the electrically driven train is connected to power station through overhead lines which carry Alternating Current. Because such current continuously changes direction and strength, the magnetic field it produces in surrounding space also changes in direction and strength. In our body, this produces eddy currents, but it is too small for a person to notice. However, if the person touches a large conducting object, such as a pipe, the current can be large and noticeable.

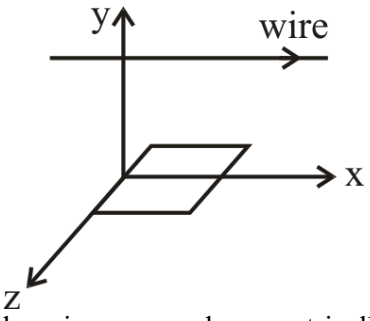
Let us try to understand this phenomena. Suppose that the man is standing vertically and his hand is touching the pipe. The overhead wire is an infinite wire carrying AC. Assume that man, pipe and the ground made a rectangular conducting loop of resistance 500Ω, dimensions as shown. The current in overhead wire is 100 sin 400t (A).



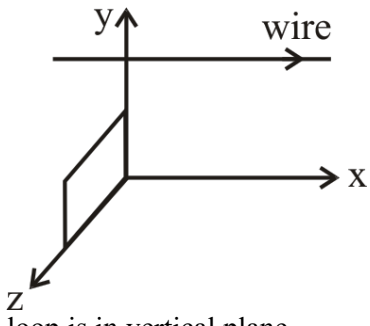
1. In which of the situations, a current will flow through the loop ?

(A)


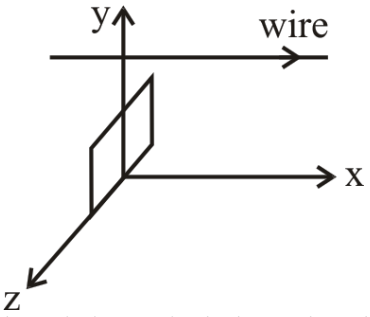
loop is on ground.

(B)


loop is on ground symmetrically placed about x-axis.

(C)


loop is in vertical plane.

(D)


loop is in vertical plane placed symmetrically about y-axis.

Ans. A

2. Select the correct statement(s) for the induced current.

(A) The induced current will have the same phase as that of over head wire.

(B) The induced current will have a maximum magnitude of $8\ell n2\ \mu A$.

(C) The induced current will have phase difference of 90° with over head wire.

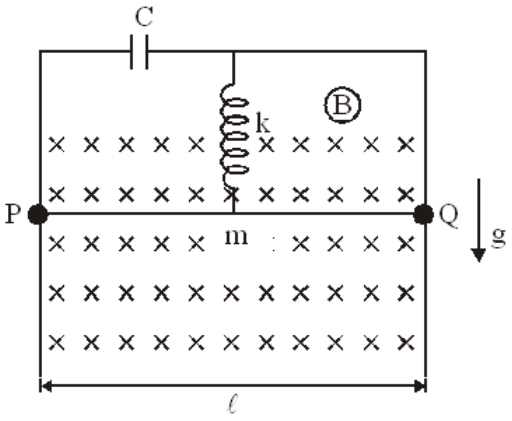
(D) The maximum magnitude of induced current is $4\ell n2\ \mu A$.

Ans. B, C

SPACE FOR ROUGH WORK

Paragraph for Questions 3 and 4

A conducting slider PQ of mass m and length ℓ is free to translate on two vertical smooth rails in the presence of gravity. The slider is connected with an initially relaxed, non conducting spring of spring constant k in uniform magnetic field B into the plane. If the slider is released from rest and capacitor is connected in circuit then



3. Motion of rod will be

(A) SHM with angular frequency $\sqrt{\frac{k}{m}}$

(B) Oscillatory but not SHM

(C) SHM with angular frequency $\sqrt{\frac{k}{m+B^2\ell^2C}}$

(D) Amplitude of oscillation of rod will be $\frac{mg}{k}$

Ans. C, D

4. If suddenly spring is cut while rod is in motion, then :

(A) the rod will still perform oscillatory motion.

(B) the rod will move downward and aquire terminal velocity.

(C) acceleration of rod will be constant.

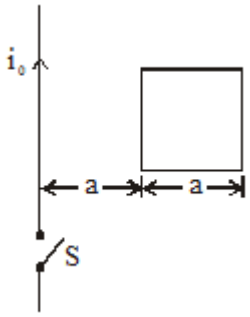
(D) charge on capacitor will increase with time.

Ans. C, D

SPACE FOR ROUGH WORK

Paragraph for Questions 5 and 6

Consider a square loop of wire with dimension a. One side of which is parallel to another long straight wire carrying current i_0, at a distance a. The resistance of the loop is R. Neglect the self inductance of the loop. Initially switch S was closed and at t = 0 it is opened.



5. What is the net amount of charge that will flow after t = 0 through a fixed cross-section of the wire of the loop?

(A) $\frac{\mu_0 i_0 \ell n(2)\, a}{\pi R}$

(B) $\frac{2\mu_0 i_0 \ell n(2)\, a}{\pi R}$

(C) $\frac{\mu_0 i_0 \ell n(2)\, a}{2\pi R}$

(D) $\frac{4\mu_0 i_0 \ell n(2)\, a}{\pi R}$

Ans. C

6. What is the net momentum P given to the loop during the switching-off of the current :-

(A) $\frac{\mu_0^2 i_0^2 \times \ell n(2)\, a}{2\pi^2 R}$

(B) $\frac{\mu_0^2 i_0^2 \times \ell n(2)\, a}{4\pi^2 R}$

(C) $\frac{\mu_0^2 i_0^2 \times \ell n(2)\, a}{\pi^2 R}$

(D) None

Ans. D

SPACE FOR ROUGH WORK

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;
Zero Marks : 0 In all other cases.

Paragraph for Questions 7 and 8

A simple pendulum of length $L = \frac{5}{2}$m has a bob of mass m = 1kg. The bob is connected to light horizontal spring of force constant k = 5 N/m. The spring is relaxed when pendulum is vertical (see fig 1.) Now consider a similar situation in which spring is replaced with elastic cord of same force constant k. The cord is relaxed when the pendulum is vertical (see fig. 2). The bob is pulled slightly in same plane and released in both cases. (Use $\pi = 3.14$, $g = 10\ m/s^2$)

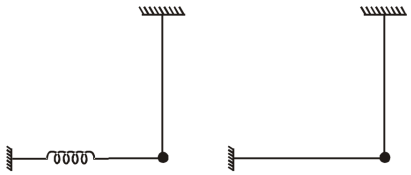


Figure-1 Figure-2

7. Time period (in sec.) of small oscillations in Fig. 1_____

Ans. 2.09

Sol. When the bob is displaced slightly by angle θ.

$\tau = mg\sin\theta\ell + kx\ell = mg\sin\theta\ell + k\ell^2\theta$ $(x = \ell\theta)$

$$m\ell^2\frac{d^2\theta}{dt^2} = -(mg\ell + k\ell^2)\theta$$

$$\frac{d^2\theta}{dt^2} = -\left(\frac{g}{\ell} + \frac{k}{m}\right)\theta$$

$$\omega = \sqrt{\left(\frac{g}{\ell} + \frac{k}{m}\right)} = 3$$

$$T = \frac{2\pi}{\omega}$$

8. Time period (in sec.) of small oscillations in Fig. 2_____

Ans. 2.61-2.62

Sol. In this case the cord is similar to spring when it is stretched but it will exert no force during half the motion when it is loose.

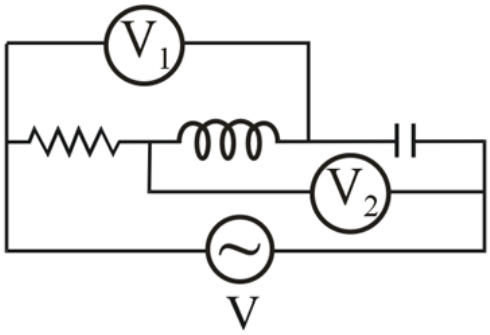
$$T = \frac{T_1}{2} + \frac{T_2}{2} = \frac{\pi}{\omega_1} + \pi\sqrt{\frac{L}{g}}$$

$$= \frac{\pi}{3} + \frac{\pi}{2} = \pi\left(\frac{5}{6}\right) = 2.62$$

SPACE FOR ROUGH WORK

Paragraph for Questions 9 and 10

In the RLC series circuit shown, the readings of voltmeters are $V_1 = 150$ V, $V_2 = 50$ V and the source has emf 130V.



9. The power factor of the circuit is $\frac{6n}{13}$ find n.

Ans. 2.00

Sol. $(V_L - V_C) = 50$

$V_R^2 + (V_L - V_C)^2 = 130^2$

$\Rightarrow V_R^2 = 130^2 - 50^2$

$\therefore$ Power factor $= \frac{V_R}{V} = \frac{120}{130} = \frac{12}{13}$

10. Value of $\frac{4V_L}{V_C}$ is:

Ans. 9.00

Sol. $V_R^2 + V_L^2 = 150^2$

$\Rightarrow V_L = \sqrt{150^2 - 120^2} = 90V$

$V_L - V_C = 50$

$\Rightarrow V_C = 90 - 50 = 40$ V

SPACE FOR ROUGH WORK

Paragraph for Questions 11 and 12

Two strings AB and BC of length 1 meter and 1/2 meter respectively are joined together at end B as shown. Point B is the junction of both strings. The mass per unit length of string AB and BC are 0.16 kg/m and 0.25 kg/m respectively. A tension of 100 N is maintained in both the strings. A sinusoidal transverse wave pulse of amplitude 2mm is produced at end 'A'. It is partly reflected and partly transmitted at joint B.



11. Find value of $\frac{A_t}{A_i}$ (where $A_t \rightarrow$ Amplitude of transmitted wave, $A_i \rightarrow$ amplitude of incidence wave)

Ans. 0.88 - 0.89

Sol. $\frac{A_t}{A_i} \Rightarrow \frac{2\sqrt{\mu_1}}{\sqrt{\mu_1}+\sqrt{\mu_2}} \Rightarrow 2\left[\frac{\sqrt{0.16}}{\sqrt{0.16}+\sqrt{0.25}}\right]$

$= \frac{2\times 0.4}{0.9} \Rightarrow \frac{0.8}{0.9} = \frac{8}{9}$

$\frac{\text{Power transmitted}}{\text{Power incident}} \Rightarrow \frac{\left(\sqrt{\mu_2}\right)(A_t)^2}{\sqrt{\mu_1}(A_i)^2}$

$\Rightarrow \frac{5}{4}\times\left(\frac{8}{9}\right)^2 \Rightarrow 0.99$

12. Calculate the value of $\frac{P_{transmitted}}{P_{incident}}$ (where $P_{transmitted} \rightarrow$ power transmitted, $P_{incident} \rightarrow$ power incident)

Ans. 0.98 - 0.99

Sol. $\frac{A_t}{A_i} \Rightarrow \frac{2\sqrt{\mu_1}}{\sqrt{\mu_1}+\sqrt{\mu_2}} \Rightarrow 2\left[\frac{\sqrt{0.16}}{\sqrt{0.16}+\sqrt{0.25}}\right]$

$= \frac{2\times 0.4}{0.9} \Rightarrow \frac{0.8}{0.9} = \frac{8}{9}$

$\frac{\text{Power transmitted}}{\text{Power incident}} \Rightarrow \frac{\left(\sqrt{\mu_2}\right)(A_t)^2}{\sqrt{\mu_1}(A_i)^2}$

$\Rightarrow \frac{5}{4}\times\left(\frac{8}{9}\right)^2 \Rightarrow 0.99$

SPACE FOR ROUGH WORK

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.
Zero Marks : 0 In all other cases

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1. A transverse periodic wave on a string with linear mass density of 0.2 kg/m is described by equation.
$y = 0.01 \sin \{220\, t - 11x\}$
where x and y are in metres and t in seconds. The tension (in N) in the string is equal to :
Ans. 80

Sol. $v = \sqrt{\frac{T}{\mu}} = \frac{\omega}{k}$

$T = \frac{\omega^2 \mu}{k^2} = \left\{\frac{220}{11}\right\}^2 \times 0.2$

$= 202 \times 0.2 \ = 400 \times 0.2 = 80$ N

2. A sinusoidal voltage of peak value 250 V is applied to a series LCR circuit, in which R = 8Ω, L = 24 mH and C = 60μF. The value of power dissipated at resonant condition is 'x' kW. The value of x to the nearest integer is _____.
Ans. 4

Sol. At resonance power (P)

$P = \frac{(V_{rms})^2}{R}$

$P = \frac{(250/\sqrt{2})^2}{8} = 3906.25$ W

≈ 4 kW

3. A linear harmonic oscillator is subjected to a periodic force such that it's amplitude is maximum at an angular frequency of 10 rad/s. Phase difference between applied force and it's velocity at this frequency is ϕ. It's damping coefficient is 0.1 / sec. Fill the value of $|\cot \phi|$.
Ans. 100

Sol. $\tan \phi = \frac{\omega L - \frac{1}{\omega C}}{R} = \frac{\omega^2 LC - 1}{\omega RC}$

$\omega = \sqrt{\omega_0^2 - 2r^2} = \sqrt{\frac{1}{LC} - \frac{R^2}{2L^2}}$

$\tan \phi = \frac{\omega^2 LC - 1}{\omega RC} = -\frac{\frac{R^2 C}{2L}}{\omega RC} = -\frac{R}{2L\omega}$

$\frac{R}{2L} = r$; $\tan \phi = -\frac{r}{\omega}$

$\tan \phi = -\frac{0.1}{10} = 0.1 = 0.01$

$\cot \phi = 100$

SECTION-IV : (Maximum Marks: 16)

- This section contains **TWO (02)** questions.
- Each question contains two columns, **Column-I** and **Column-II**.
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **five** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **Column-II**.
- One or more entries in **Column-I** may match with one or more entries in **Column-II**.
- For each entry in **Column-I**, darken the options of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then chosen these three options. Similarly, for entries (B), (C) and (D).
- For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**

Full Marks	:	+2	If only the option(s) corresponding to all the correct match(es) is (are) chosen
Zero Marks	:	0	If none of the options is chosen
Negative Marks	:	–1	In all other cases

SPACE FOR ROUGH WORK

1. A series LCR circuit is connected to a 45 sin (ωt) Volt source. The resonant angular frequency of the circuit is 10^5 rad s^{-1} and current amplitude at resonance is I_0. When the angular frequency of the source is $\omega = 8 \times 10^4$ rad s^{-1}, the current amplitude in the circuit is 0.05 I_0. If L = 50 mH, match each entry in column-I with an appropriate value from column-II and choose the correct option.

Column-I		Column-II	
(A)	I_0 in mA	(P)	44.4
(B)	The quality factor of the circuit	(Q)	18
(C)	The bandwidth of the circuit in rad s^{-1}	(R)	400
(D)	The peak power dissipated at resonance in Watt	(S)	2250
		(T)	500

Ans. A ⟶ Q, B ⟶ R, C ⟶ T, D ⟶ P

Sol. $V = 45 \sin \omega t, \quad L = 50$ mH

$$\omega_0 = 10^5 \text{rad/s} = \frac{1}{\sqrt{LC}}$$

$$\Rightarrow C = \frac{1}{L\omega_0^2} = \frac{1}{5 \times 10^{-2} \times 10^{10}} = 2 \times 10^{-9} \text{ F}$$

$$I_0 = \frac{45}{R}$$

$$\omega = 8 \times 10^4 \text{ rad/s} = 0.8\ \omega_0$$

$$I = 0.05 I_0 = \frac{I_0}{20} \Rightarrow Z = 20R$$

$$X_L = 8 \times 10^4 \times 5 \times 10^{-2}\,\Omega = 4k\Omega$$

$$X_C = \frac{1}{8 \times 10^4 \times 2 \times 10^{-9}} = \frac{1}{16} \times 10^5\,\Omega = \frac{25}{4}k\Omega$$

$$Z^2 = R^2 + (X_C - X_L)^2$$

$$400R^2 = R^2 + \left(\frac{9}{4}k\Omega\right)^2$$

$$R = \frac{\frac{9}{4}k\Omega}{\sqrt{399}} \approx \frac{9}{80}k\Omega = \frac{900}{8}\Omega$$

$$I_0 = \frac{V_0}{R} = \frac{45 \times 8}{900} = \frac{8}{20}A \approx 0.4A = 400mA$$

$$Q = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{8}{900}\sqrt{\frac{5 \times 10^{-2}}{2 \times 10^{-9}}} = \frac{8}{900}\sqrt{25 \times 10^6}$$

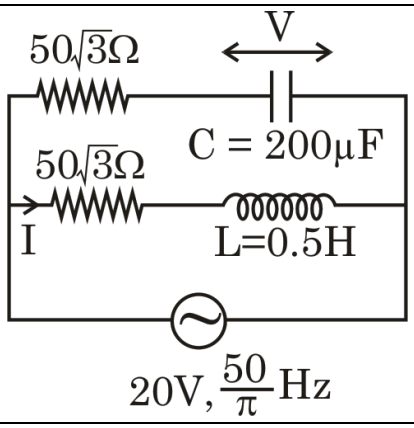
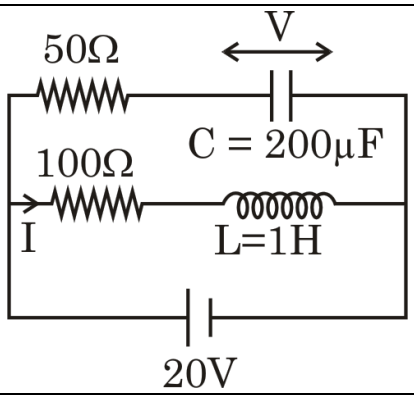
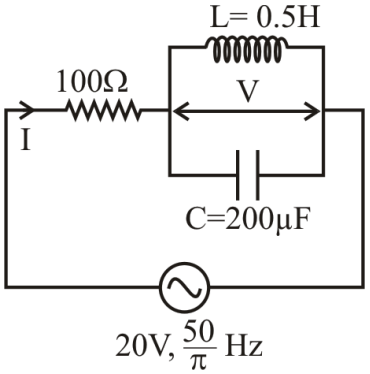
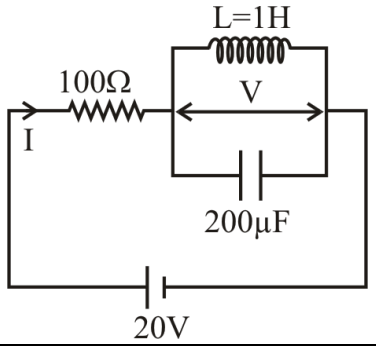
$$Q = \frac{8}{900} \times 5000 = 44.4$$

$$Q = \frac{\omega_0}{\Delta\omega} \Rightarrow \Delta\omega = \frac{\omega_0}{Q} = \frac{10^5}{44.4} = 2250.0$$

$$P_{max} = I_0^2 R = \frac{45^2}{R^2} \times R = \frac{45^2}{R} = \frac{45^2}{900} \times 8 = 18.4W$$

SPACE FOR ROUGH WORK

2. Here I and V represent current and voltage respectively (steady state for DC and rms for AC) match Column-I with Column-II using given codes.

Column-I		Column-II	
(A)		(P)	I = 0.2 A
(B)		(Q)	I = 0 A
(C)		(R)	V = 20 Volt
(D)		(S)	V = 0 Volt
		(T)	V = 10 Volt

Ans. A → P, A → T, B → P, B → R, C → Q, C → R, D → P, D → S

SPACE FOR ROUGH WORK